

WHAT IS CTF (CAPTURE THE FLAG)?

A Detailed Report with Recent Case Studies (2025–2026)

1. Definition

Capture The Flag (CTF) is a practical cybersecurity competition in which participants solve security challenges to discover hidden strings called flags. CTFs provide controlled environments for developing skills in penetration testing, network security, digital forensics, cryptography, reverse engineering, and incident investigation.

Competitions range from beginner-friendly educational events to advanced international competitions such as DEF CON CTF. CTFtime helps participants discover events and track results and team rankings. For aspiring SOC Analysts, CTFs offer practice in technical investigation, evidence analysis, recognizing suspicious activity, and communicating findings.

2. Objectives of CTF Competitions

- Develop practical skills by applying cybersecurity concepts to hands-on challenges.
- Improve problem-solving through structured investigation of unfamiliar technical problems.
- Understand attack techniques in authorized and controlled environments.
- Build teamwork through task delegation, collaboration, and sharing findings.
- Prepare for cybersecurity careers in SOC operations, penetration testing, forensics, and security research.

3. Main Competitions Format

Format	How It Works
Jeopardy-style	Participants choose independent challenges from categories such as web exploitation, cryptography, forensics, and reverse engineering. Each challenge awards points according to difficulty. Example: analyze an image containing hidden data and recover the flag.
Attack-Defense (A/D)	Teams maintain and defend their own vulnerable services while attempting to exploit services belonging to other teams, within competition rules. Example: identify and patch a vulnerability in a service while monitoring it for attacks.
King of the Hill (KoH)	Teams compete to gain and hold control of a shared system or service over time.

4. Common Challenge Categories

- Web exploitation — finding and abusing vulnerabilities in web applications
- Pwn / binary exploitation — exploiting memory-corruption bugs in compiled programs
- Reverse engineering — analyzing compiled binaries or firmware
- Cryptography — breaking weak or misconfigured cryptographic schemes
- Forensics — recovering evidence from disk, memory, or network captures
- OSINT / Misc — open-source intelligence and miscellaneous puzzle-style tasks
- Hardware / IoT / embedded — increasingly common at DEF CON villages and MITRE's eCTF

5. Major CTF Platforms and Events

Platform / Event	Main purpose	Typical audience
picoCTF	Educational cybersecurity challenges	Beginners and students
DEF CON CTF	Advanced technical competition	Experienced researchers
CTFtime	Event calendar, results, rankings	CTF participants
Vendor/company CTFs	Product- or skill-focused challenges	Students and professionals
Conference CTFs	Hands-on activities alongside conferences	Researchers and practitioners

CTFtime is a tracking and discovery platform rather than a competition itself. DEF CON CTF is an advanced event associated with the DEF CON conference, while picoCTF focuses on accessible cybersecurity education.

Why Organizations Run CTFs

- Skills training for students and professionals in a hands-on, low-stakes setting
- Recruitment and talent pipelines for companies, universities, and governments
- Community and diplomacy building among security teams across organizations or countries
- Research and benchmarking, including — recently — testing how well AI systems solve security tasks

6. Recent Case Studies (2025–2026)

Case Study 1 — DEF CON CTF 2026 — "Blue Water" Wins the World Championship

DEF CON CTF is often called the "hacking Olympics." It's an international hacking competition where the world's top white-hat hackers compete in vulnerability analysis, attack-and-defense, and problem-solving; this year's qualifying round drew 686 teams worldwide, of which only 12 advanced to the Las Vegas finals.

The finals (DEF CON 34, August 6–9, 2026) were won by Blue Water, an international joint team combining engineers and students from multiple countries — including GMO Internet Group's Japanese security units and Tsinghua University's Blue Lotus team from China. Blue Water took first place after finishing second for three consecutive years. The margin was dramatic: Blue Water finished with roughly 578,000 points, more than 120,000 points ahead of the second-place team.

This case illustrates two trends: (1) elite CTF play is now built on multi-national, cross-company alliances rather than single-organization teams, and (2) organizational succession matters — in 2026, DEF CON selected a new group, the Benevolent Bureau of Birds, to organize the competition for a four-year term, formed from members of Carnegie Mellon's Plaid Parliament of Pwning, the University of British Columbia's Maple Bacon, and a team called The Duck. Those same three groups had previously competed together as Maple Mallard Magistrates, winning four consecutive DEF CON CTF titles from 2022–2025.

Case Study 2 — OWASP Secure Development CTF at DEF CON 34 (2026)

OWASP's 2026 event used deliberately vulnerable applications to teach participants to find and repair software vulnerabilities. The event described 321 challenges across six applications, including Juice Shop, DVWA, WebGoat, and Security Shepherd. Participants identified issues, patched them, and submitted GitHub pull requests; automated checks validated fixes.

Learning points: CTFs can teach remediation and secure coding, not only exploitation. SOC relevance: understanding vulnerability classes and fixes helps analysts communicate findings to engineering teams.

Case Study 3 — NCIIPC–AICTE Pentathlon 2026 (India)

The 2026 Pentathlon included an online Jeopardy-style CTF as an initial stage for identifying cybersecurity and penetration-testing talent. More than 200 finalists advanced to a practical bootcamp and finale at Manipal Institute of Technology. Topics included cloud security, malware analysis, cryptography, incident response, and vulnerability assessment.

Learning points: CTFs can lead into more complex team scenarios and cover both offensive and defensive topics. SOC relevance: incident response and malware analysis map to common SOC investigations.

Case Study 4 — HITCON Cyber Range Defensive CTF (2025)

HITCON's 2025 Cyber Range competition focused on defensive skills. Its qualifier involved firewall, Windows, Linux, and endpoint logs to identify suspicious activity and reconstruct attack paths. Final exercises included incident investigation, operational response, configuration changes, rule writing, and defensive improvements.

Learning points: log correlation and attack-chain reconstruction are central to blue-team exercises. SOC relevance: the workflow resembles alert triage, evidence collection, IOC identification, and containment.

Case Study 5 — Cross-Border Collaboration — GÉANT Security Days

CTFs are also used diplomatically, to build trust between national research-and-education networks. At Security Days 2026, a CTF designed by Brazil's national research network RNP brought together research-and-education network teams from across Europe and beyond, demonstrating how gamified environments bridge technical skill-building with strategic thinking. The winning team came from Greece's GRNET, whose CISO attributed the win to sustained, ongoing training rather than short-term cramming, reinforcing that consistent practice in realistic environments is more valuable than short bursts of preparation.

Case Study 6 — Academic and Community CTFs

Smaller, local CTFs remain a major training vector for students. At BSidesROC 2026 in Rochester, NY, a Gannon University cybersecurity student team took fourth place among more than 20 teams in a continuous six-hour CTF, part of a newly formed campus cyber club built around weekly hands-on training sessions. Separately, East Carolina University ran its first "Cyber Capture the Flag" competition in January 2026 with support from the North Carolina National Guard, whose adjutant general framed it around the everyday cyber threats facing banking, insurance, and healthcare. These cases show CTFs functioning as a grassroots on-ramp into the profession, often with government or military sponsorship.

Case Study 7 — Industry–Nonprofit Youth Programs

CTFs are also used to build the future talent pool at the K-12/early-career level. Canada's ICTC runs its CyberTitan youth initiative on the Field Effect Cyber Range, with a 2026 scenario built around a maple-syrup factory under active cyberattack that students must contain while keeping operations running — explicitly designed to draw new entrants into the security talent pipeline.

Case Study 8 — Geopolitics — State-Sponsored CTF Ecosystems

CTFs have become a policy topic, not just a technical one. An Atlantic Council report, "Capture the (red) flag," examined China's use of hacking competitions as a state talent pipeline. The report argues Western governments should sponsor more capture-the-flag competitions to better cultivate domestic cybersecurity hacking skills, taking a page from what it describes as China's world-leading hacking-contest ecosystem, which has been used to help develop the country's offensive hacking talent. This

case shows CTFs being discussed at the policy level as an instrument of national competitiveness, not just an educational tool.

Case Study 9 — The AI Disruption Question

The most consequential emerging trend is large language models' growing ability to solve CTF challenges. A 2026 academic study frames this directly: CTF competitions are among cybersecurity's most effective training grounds for skills in cryptography, web exploitation, and binary exploitation, but LLMs can now solve a growing share of challenges with minimal human input, raising urgent questions about fairness, the validity of rankings, and whether participation still delivers the learning that justifies the effort. The study combines a synthesis of published benchmarks — including a recent government evaluation — case studies of live competitions across three challenge categories, observation of community debate, and interviews with experienced players and organizers. This connects directly to Case Study 2 (Raonsecure), where AI-assisted solving was treated as a competitive edge rather than a fairness problem — showing the community is still working out where the line should be drawn.

7. Common Tools Used in CTFs

Tool	Purpose	Example application
Wireshark	Network traffic analysis	Inspect DNS or HTTP traffic
Burp Suite	Web security testing	Investigate web vulnerabilities
Nmap	Network discovery and scanning	Identify hosts and ports in a lab
Kali Linux	Security testing environment	Run authorized security tools
CyberChef	Data transformation and decoding	Decode and inspect encoded strings
Volatility	Memory forensics	Investigate processes in memory dumps
Python	Scripting and automation	Parse logs and automate analysis

8. CTF Activities mapped to SOC Responsibilities

CTF activity	Related SOC task
Analyze packet captures	Investigate suspicious network connections
Examine event logs	Correlate SIEM alerts and system activity
Identify malicious files	Analyze malware indicators
Reconstruct an attack path	Build an incident timeline
Identify vulnerabilities	Support vulnerability management
Write scripts	Automate alert enrichment and log analysis
Explain the solution	Document findings and communicate incidents

9. Recommendations for an Aspiring SOC Analyst

- Start with beginner challenges in networking, Linux, and digital forensics.
- Practice interpreting logs and network evidence, not only offensive exploitation.
- Map relevant findings to MITRE ATT&CK when appropriate.
- Document each challenge: objective, tools, evidence, analysis, and lessons learned.
- Reproduce selected investigations in a controlled lab using Microsoft Sentinel or another SIEM.
- Join team competitions to practice communication and task delegation.

10. Outlook

CTFs provide structured, hands-on cybersecurity practice, from introductory educational challenges to advanced international competitions and defensive cyber-range exercises. Recent examples show how competitions can cover vulnerability discovery, remediation, incident investigation, and team-based defense.

For aspiring SOC Analysts, the most useful outcome is not simply collecting flags or achieving a high rank. It is learning to investigate evidence systematically, recognize suspicious behavior, understand technical findings, and explain what happened in a clear and reproducible way.